

Lab 5:

Objective

To design and implement wishlist feature.

Theory

A wishlist feature allows users to save items they are interested in for later viewing, without committing to a bid immediately. This is a common feature in e-commerce and auction platforms, as it lets users track multiple items of interest across browsing sessions and revisit them easily from a dedicated page.

In this system, a separate wishlist table was created to store the relationship between users and the items they have saved. Each row links a user_id to an item_id, and a unique constraint on this pair ensures that the same user cannot wishlist the same item more than once. When a user clicks the wishlist button on an item, the system checks whether a matching row already exists in the wishlist table — if it does, the row is deleted (removing the item from the wishlist); if it does not, a new row is inserted (adding the item to the wishlist). This toggle behavior allows a single button to handle both adding and removing an item.

To improve user experience, the wishlist toggle was implemented using JavaScript's fetch() function to send the request to the server in the background, rather than submitting a traditional HTML form. This allows the button's text and icon to update instantly without reloading the page, since the PHP script responds with a JSON result indicating the new wishlist state instead of redirecting to a new page.

Source Code:

a) Toggling Wishlist Status:

```
$check = $pdo->prepare('SELECT id FROM wishlist WHERE user_id = :user_id
AND item_id = :item_id');
$check->execute([':user_id' => $_SESSION['user_id'], ':item_id' => $itemId]);
$existing = $check->fetch();

if ($existing !== false) {
    $delete = $pdo->prepare('DELETE FROM wishlist WHERE id = :id');
    $delete->execute([':id' => $existing['id']]);
    echo json_encode(['wishlist' => false]);
} else {
    $insert = $pdo->prepare('INSERT INTO wishlist (user_id, item_id) VALUES
(:user_id, :item_id)');
    $insert->execute([':user_id' => $_SESSION['user_id'], ':item_id' => $itemId]);
    echo json_encode(['wishlist' => true]);
}
```

b) Updating the Button Instantly with JavaScript:

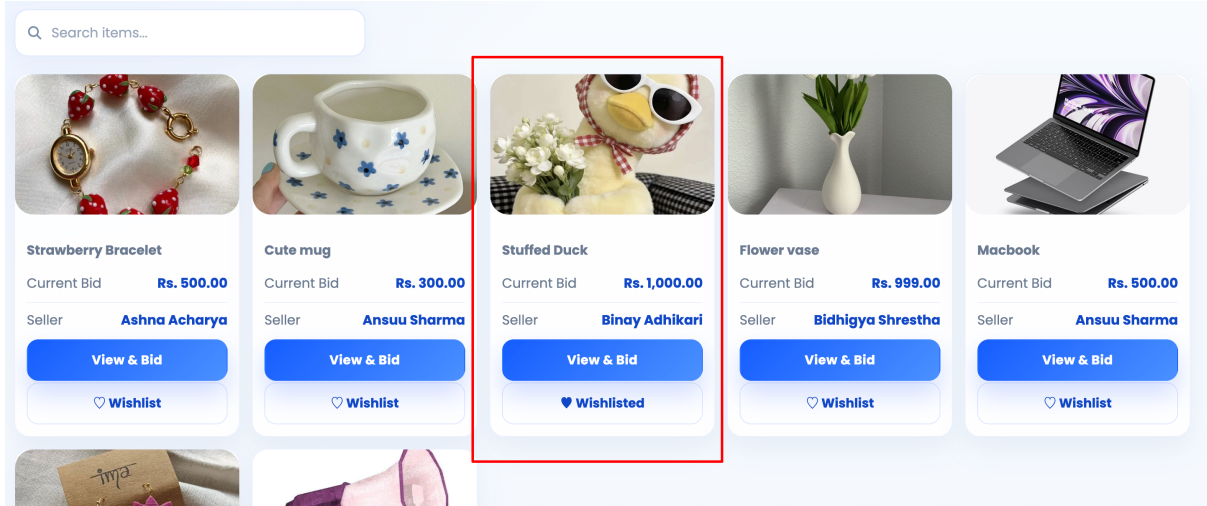
```
const response = await fetch('toggle_wishlist.php', {
    method: 'POST',
    headers: { 'Content-Type': 'application/x-www-form-urlencoded' },
    body: `item_id=${itemId}`,
});

const data = await response.json();
```

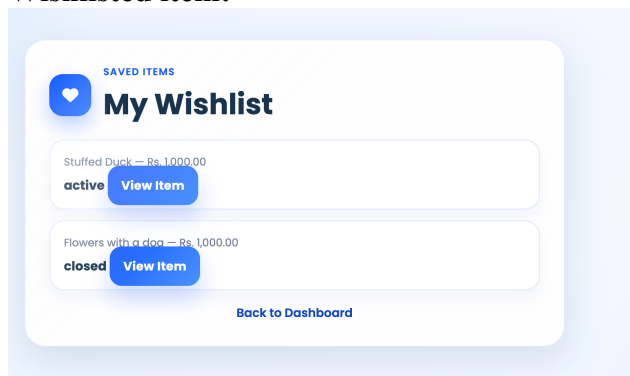
```
button.textContent = data.wishlisted ? '♥ Wishlisted' : '♡ Wishlist';
```

Output

Wishlist button:



Wishlisted item:



Conclusion

The wishlist feature was successfully implemented, allowing users to save and remove items of interest using a toggle-based system backed by a dedicated database table. The use of a unique constraint on the user-item pair ensures data integrity by preventing duplicate entries. By handling the toggle action through JavaScript's `fetch()` API and a JSON response, the feature updates instantly without a full page reload, providing a smoother user experience similar to modern web applications.